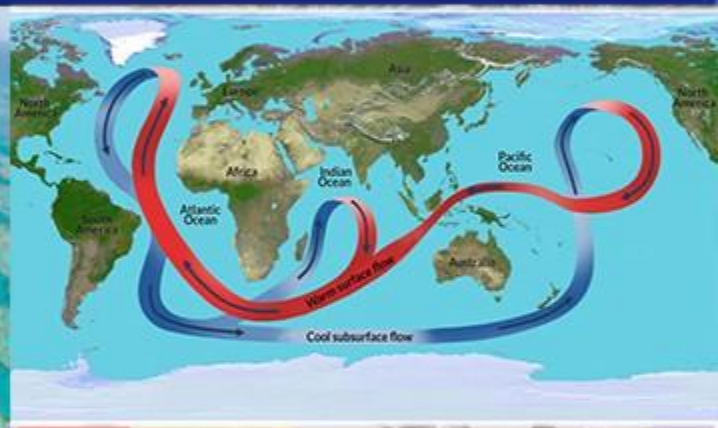


GEOGRAPHY

PHYSICAL GEOGRAPHY

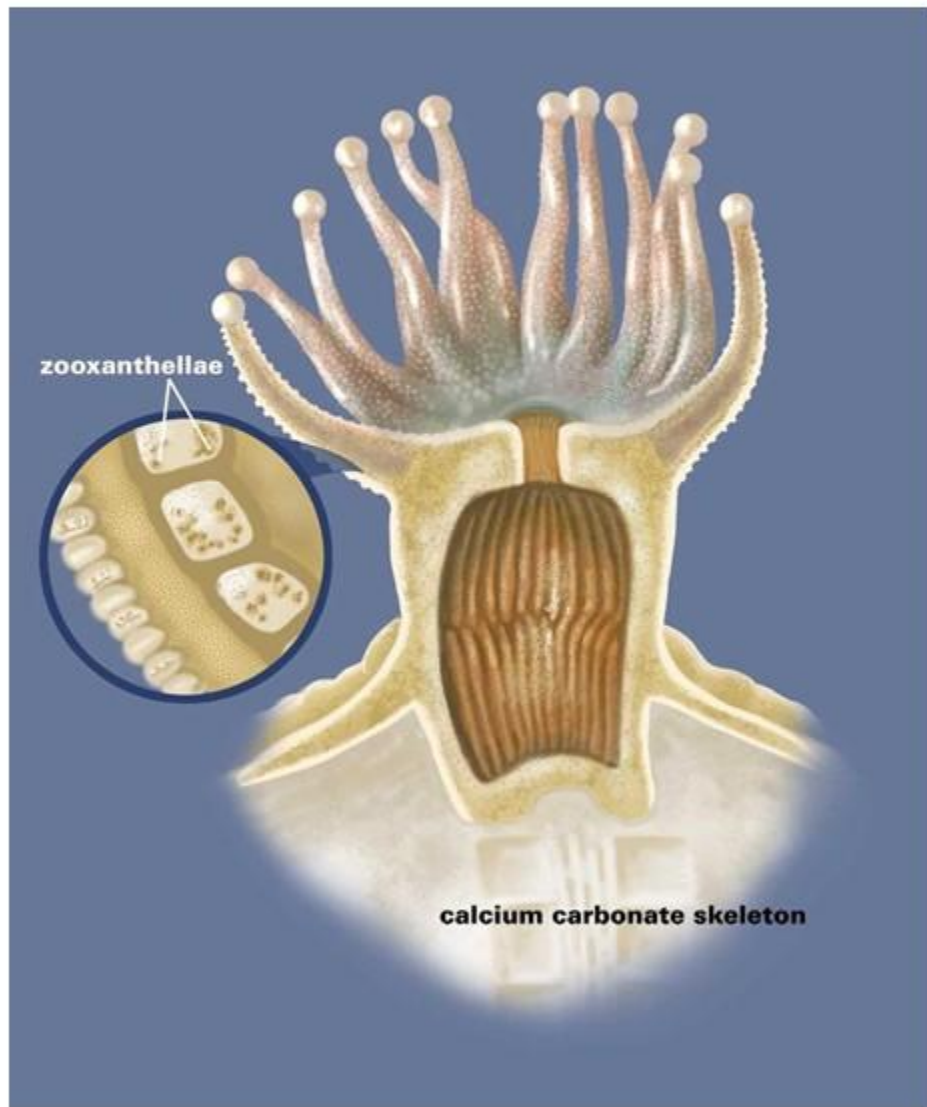


OCEANOGRAPHY

MODULE - 8

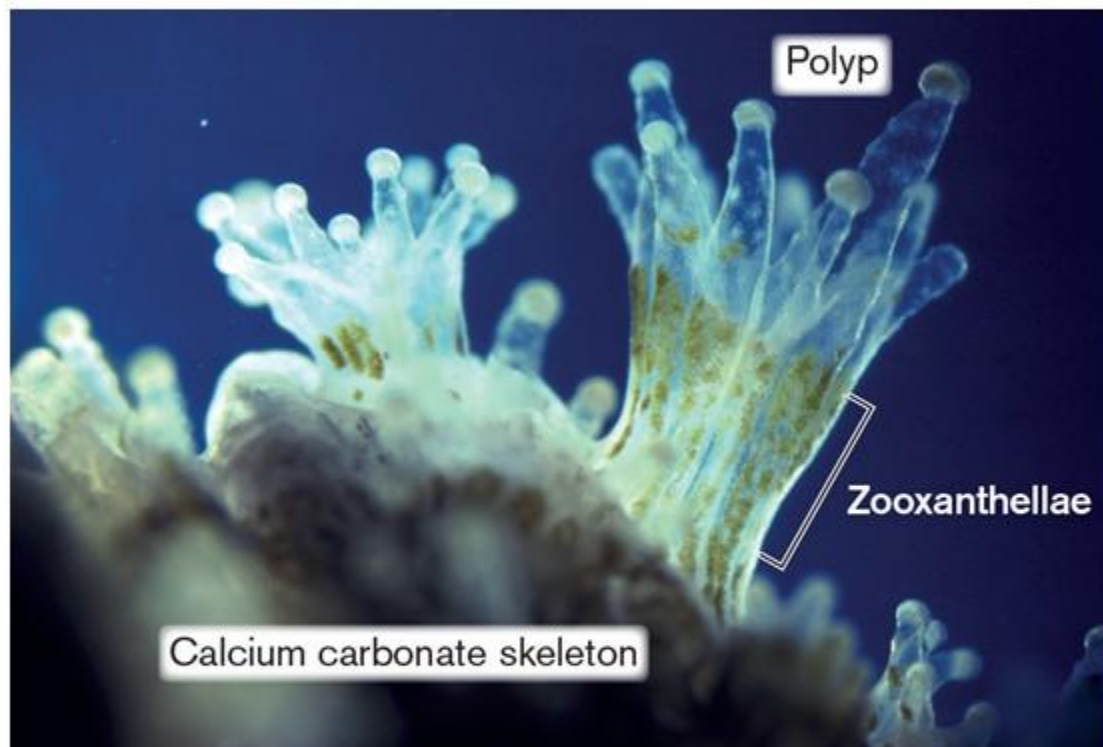
CORAL REEFS

WHAT DO YOU UNDERSTAND BY CORAL REEF?



- 1 Coral reefs are built by and made up of thousands of tiny animals—coral “polyps”—that are related to anemones and jellyfish.
- 2 Polyps are shallow water organisms which have a soft body covered by a calcareous skeleton.
- 3 The polyps extract calcium salts from sea water to form these hard skeletons.
- 4 The tubular skeletons grow upwards and outwards as a cemented calcareous rocky mass, collectively called corals.

WHAT DO YOU UNDERSTAND BY CORAL REEF?



5 When the coral polyps die, they shed their skeleton [coral] on which new polyps grow.

6 The cycle is repeated for over millions of years leading to accumulation of layers of corals [shallow rock created by these depositions is called reef].

7 These layers at different stages give rise to various marine landforms. One such important landform is called coral reef.

8 Coral reefs over a period of time transform or evolve into coral islands (Lakshadweep).

OPTIMUM CONDITIONS FOR THE
GROWTH OF CORAL REEFS

CONDITION NO. 1

OPTIMUM TEMPERATURE

- 1 As a general rule, the water temperature must not fall below 20°C.



- 2 Again they **will not flourish where there are cold currents** because of the upwelling of the cold water from the depths that cools the warm surface water.
- 3 This explains why coral reefs are generally absent on **the western coasts of continents**.
- 4 On the other hand the **warming effect of the warm currents**, e.g. the Gulf Stream, means that corals are found far to the north of the West Indies in the Atlantic Ocean.
- 5 The Pacific and the Indian Oceans, however, have the most numerous coral reefs.

OPTIMUM CONDITIONS FOR THE
GROWTH OF CORAL REEFS

CONDITION NO. 2

DEPTH OF THE OCEAN



- 1 Corals do not live in deeper waters i.e., not more than 200-250 ft. because beyond this depth **sunlight is too faint for photosynthesis to take place.**
- 2 This is essential for the survival of the **microscopic algae**, on which the coral polyps depend.
- 3 **Shallow water of less than 100 feet is ideal.** But there should always be plenty of water as polyps cannot survive for too long out of water.

OPTIMUM CONDITIONS FOR THE
GROWTH OF CORAL REEFS

CONDITION NO. 3

SEDIMENT AND SALTATION



*There should be less
sedimentation in the oceans*

- 1 The water should be saltish and free from sediment.
- 2 Corals therefore survive best in the moving ocean water well away from the silty coasts or muddy mouths of streams.
- 3 The corals are best developed on the seaward side of the reef, where constantly moving waves, tides and currents maintain an abundant supply of clear, oxygenated water.
- 4 They also bring an adequate supply of food in the form of microscopic organisms.

OPTIMUM CONDITIONS FOR THE
GROWTH OF CORAL REEFS

CONDITION NO. 3

SEDIMENT AND SALTATION

- 5 It may be pointed out that though coral polyps require sediment-free water but fresh water is also injurious for the growth of corals.



- 6 This is why corals avoid coastal lands and live away from the areas of river mouths.
- 7 Very high proportion of oceanic salinity is injurious for the growth of coral polyps because such waters contain little amount of calcium carbonates whereas lime is important food of coral polyps.
- 8 The oceanic salinity ranging between 27‰ and 30‰ is most ideal for the growth and development of coral polyps.

OCEANOGRAPHY

OPTIMUM CONDITIONS FOR THE GROWTH OF CORAL REEFS

CONDITION NO. 4

OCEAN CURRENTS AND WAVES



- 1 Ocean currents and waves are favourable for corals because they bring necessary food supply for the polyps.
- 2 It is obvious that corals grow in open seas and oceans but they die in small enclosed seas because of lack of supply of food.
- 3 Currents and waves also determines the shapes of coral reefs.

OCEANOGRAPHY

OPTIMUM CONDITIONS FOR THE GROWTH OF CORAL REEFS

CONDITION NO. 5

HUMAN ACTIVITIES AND GLOBAL WARMING



1 Human economic activities viz. deforestation, industrialization etc. causing global warming adversely affect coral in their habitats.

2 Corals are more susceptible to long-term climatic change.

3 Corals are generally termed as “rainforests” of the oceans.

4 These cannot survive in extreme warm environment .

5 The scientists claim that many of the corals have died and become skeletons due to global warming caused by anthropogenic factors mainly industrialization.

OPTIMUM CONDITIONS FOR THE GROWTH OF CORAL REEFS

CONDITION NO. 5

HUMAN ACTIVITIES AND GLOBAL WARMING



6

Besides global warming, human activities at local to regional levels such as

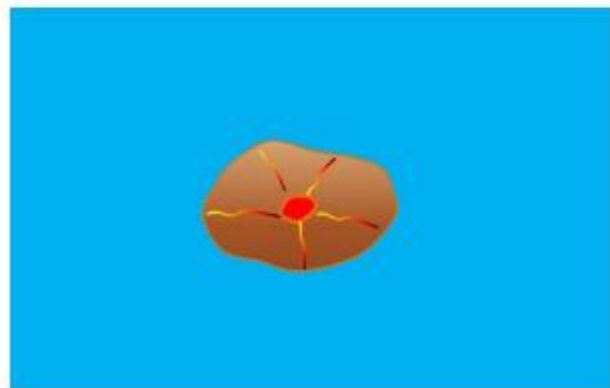
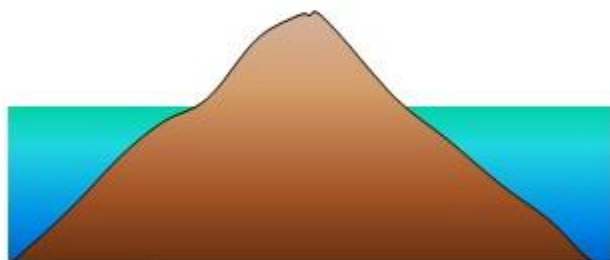
- Pollution of oceanic water through excess flux of sediments and nutrients
- Industrials effluents
- Urban wastes
- Sewage
- over fishing;
- clearances of maritime forest and filling of wetlands
- mining of coral rocks
- Collection of rare species etc. cause fatal diseases to corals.

OCEANOGRAPHY

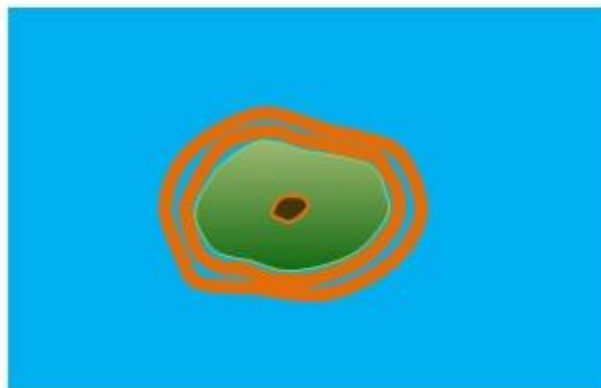
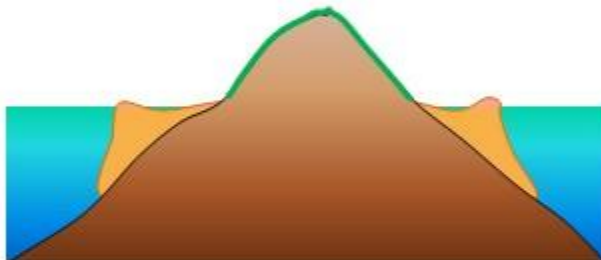
CORAL REEFS

TYPES OF CORAL REEFS

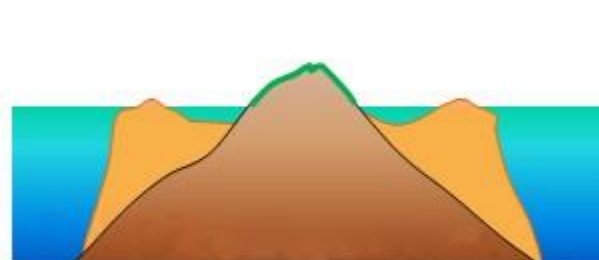
Volcanic island



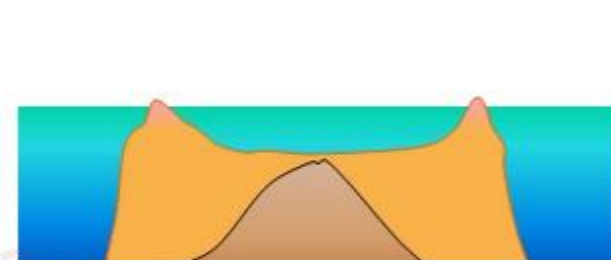
Fringing reef



Barrier reef



Atoll



OCEANOGRAPHY

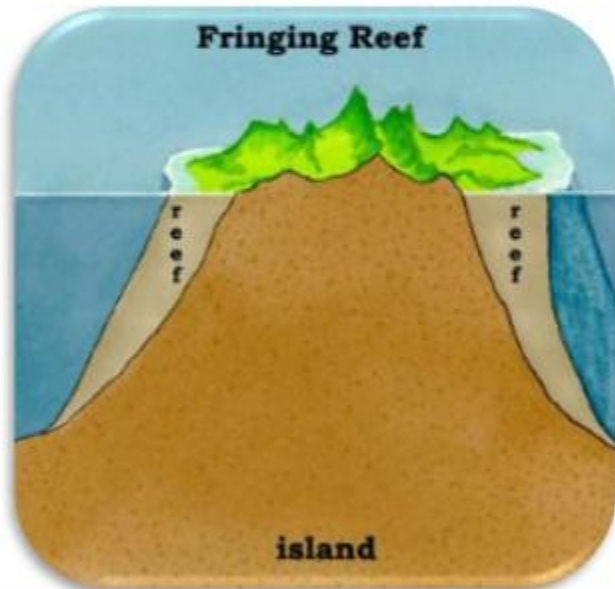
CORAL REEFS

TYPES OF CORAL REEFS

There are three main types of coral reefs.

TYPE 1

FRINGING REEFS



- 1 A fringing reef is a coralline platform lying close to the shore extending outwards from the mainland.

- 2 It is sometimes separated from the shore by a shallow lagoon.

- 3 It is widest when fringing a protruding headland but completely absent when facing the mouth of a stream.

- 4 The outer edge grows rapidly because of the splashing waves that continuously renew the supply of fresh food.

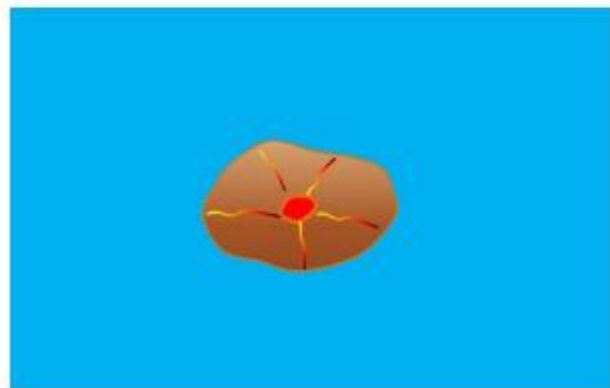
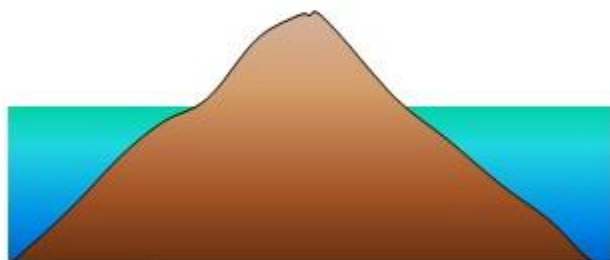
- 5 The reefs may be about a mile wide, lying just above the level of low water and sloping steeply downwards on the seaward side to a depth of about 100 feet.

OCEANOGRAPHY

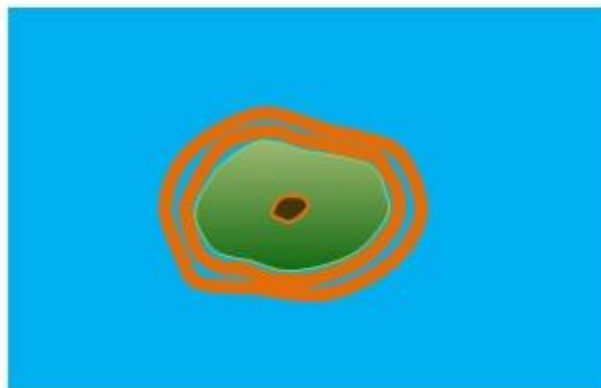
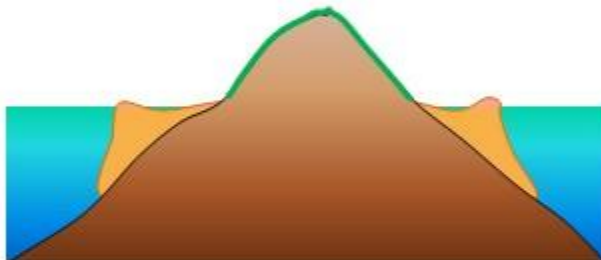
CORAL REEFS

TYPES OF CORAL REEFS

Volcanic island



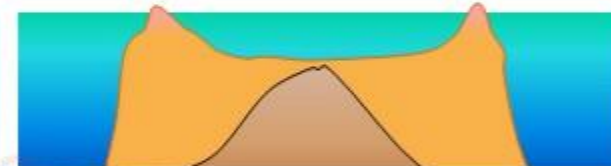
Fringing reef



Barrier reef



Atoll



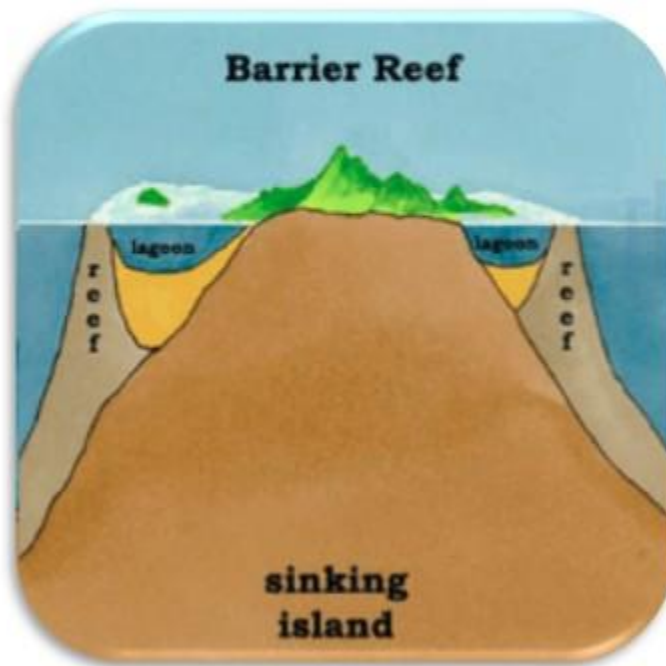
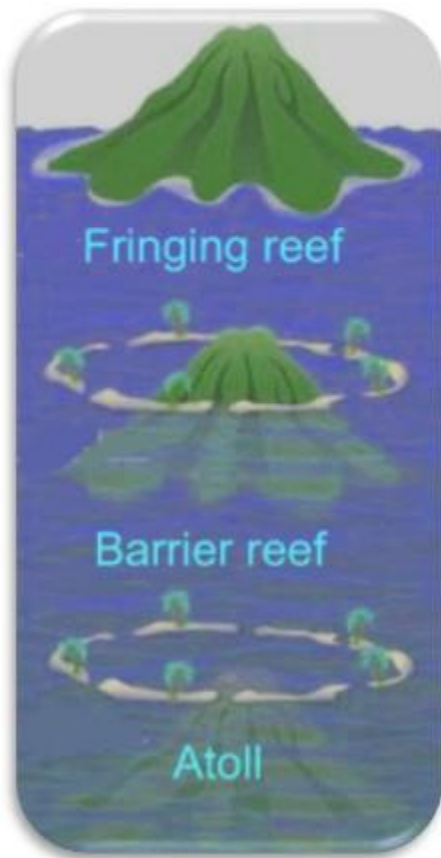
OCEANOGRAPHY

CORAL REEFS

TYPES OF CORAL REEFS

TYPE 2

BARRIER REEFS



- 1 A barrier reefs is separated from the coast by a **much wider and deeper channel or lagoon**.
- 2 The reef is partially submerged. Where it lies above the water level and sand can accumulate on it, a little vegetation is possible.
- 3 The **barrier reefs have narrow gaps** at several places to allow the water from the enclosed lagoon to return to the open ocean.

OCEANOGRAPHY

CORAL REEFS

TYPES OF CORAL REEFS

TYPE 2

BARRIER REEFS



4 Such gaps are very **useful for shipping** and provide the only entrances for ships to enter or leave the lagoon.

5 The best known barrier reef is the **Great Barrier Reef** off the coast of Queensland, Australia.

6 It is 1,200 miles long, separated from the coast by a channel 100 miles wide in places and over 200 feet deep.

OCEANOGRAPHY

CORAL REEFS

TYPES OF CORAL REEFS

TYPE 3

ATOLLS



- 1 Atolls are similar to barrier reefs except that they are **circular in shape, enclosing a shallow lagoon** without any land in the centre.
- 2 The encircling ring is usually broken in a few places to allow the free flow of water.
- 3 On the inside of the reefs, sand and limestone debris collect and palm trees like coconuts may grow.
- 4 Such palm trees thrive well in the brackish water of the lagoon.

OCEANOGRAPHY

CORAL REEFS

TYPES OF CORAL REEFS

TYPE 3

ATOLLS



5 The nuts fall into the water and are distributed widely by floating from one coral island to another.

6 The calm waters are useful for fishing and canoeing.

7 Some of the large atolls, e.g. Suvadiva in the Maldives, west of Ceylon, have a lagoon over 40 miles across.

OCEANOGRAPHY**TOPICS TO BE COVERED**

- 1. OCEAN BOTTOM RELIEF**
- 2. SALINITY AND TEMPERATURE OF OCEANS**
- 3. OCEAN TIDES**
- 4. OCEAN CURRENTS**
- 5. CORAL REEFS**

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